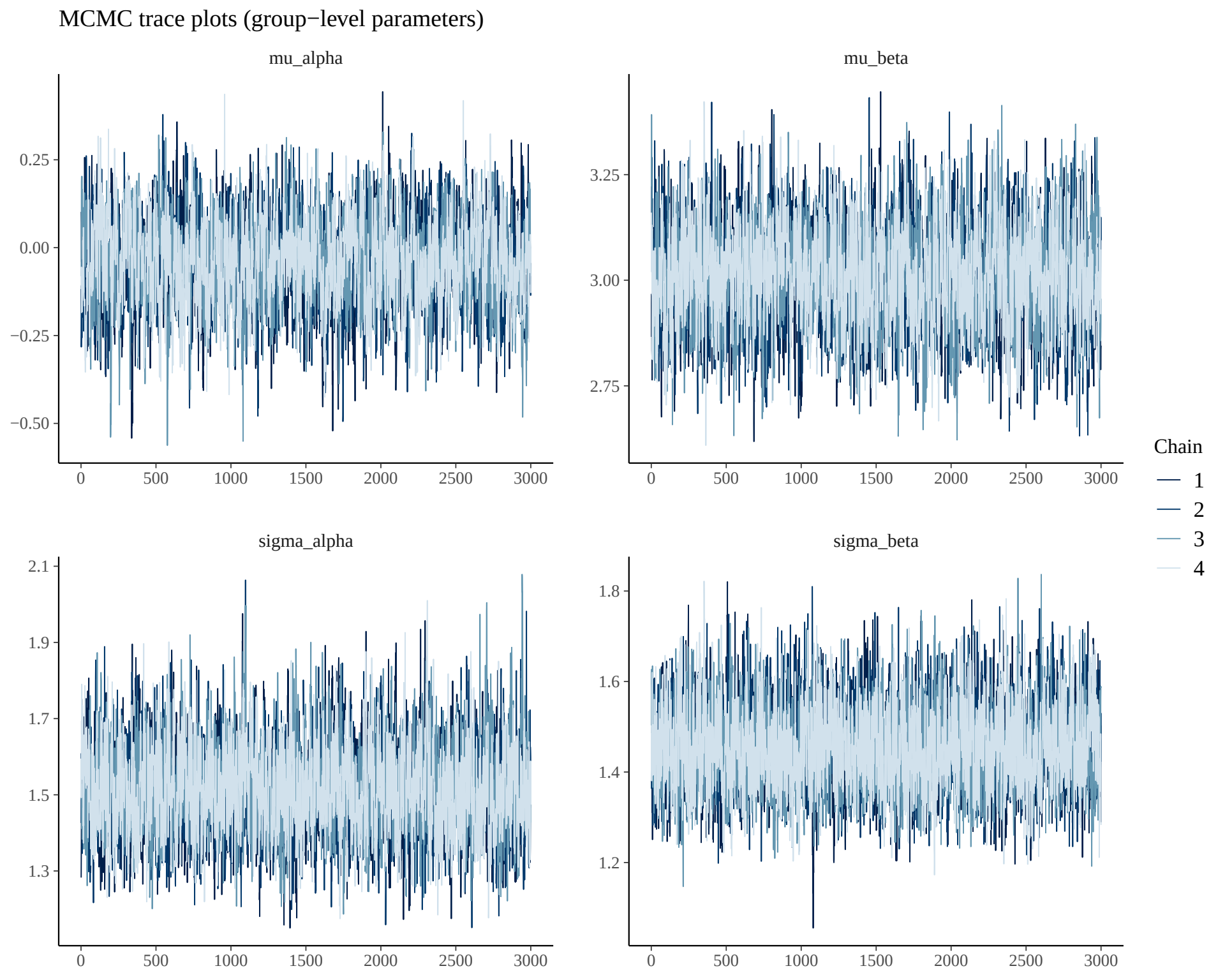
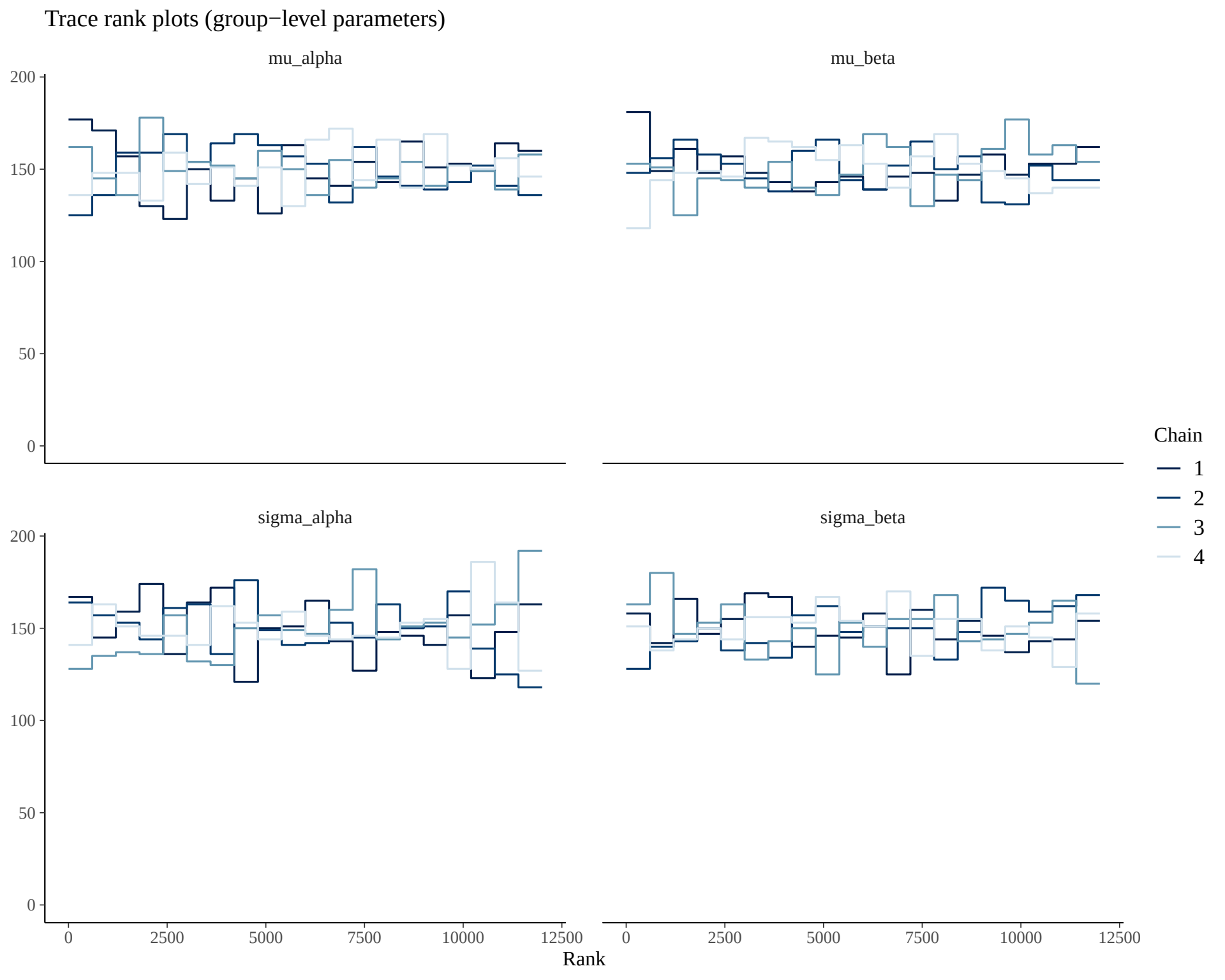
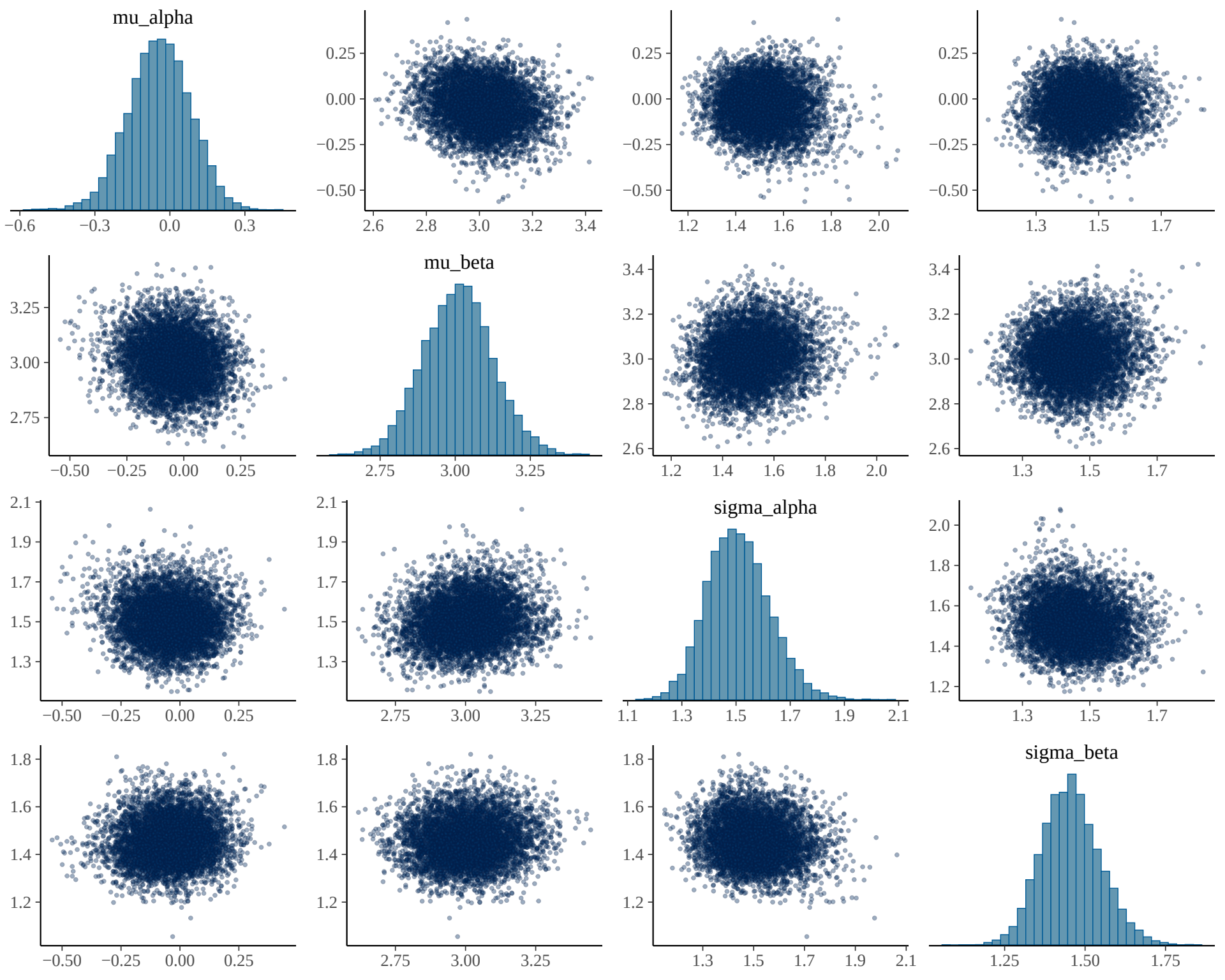
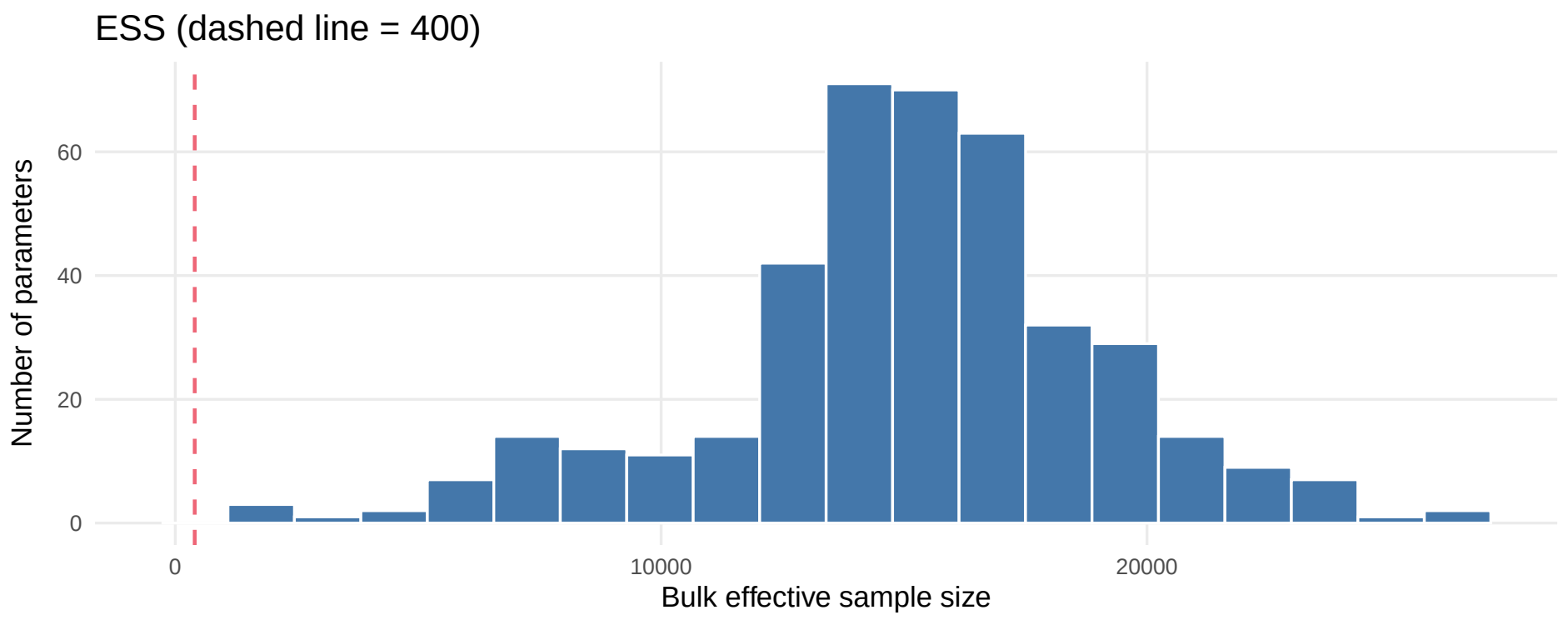
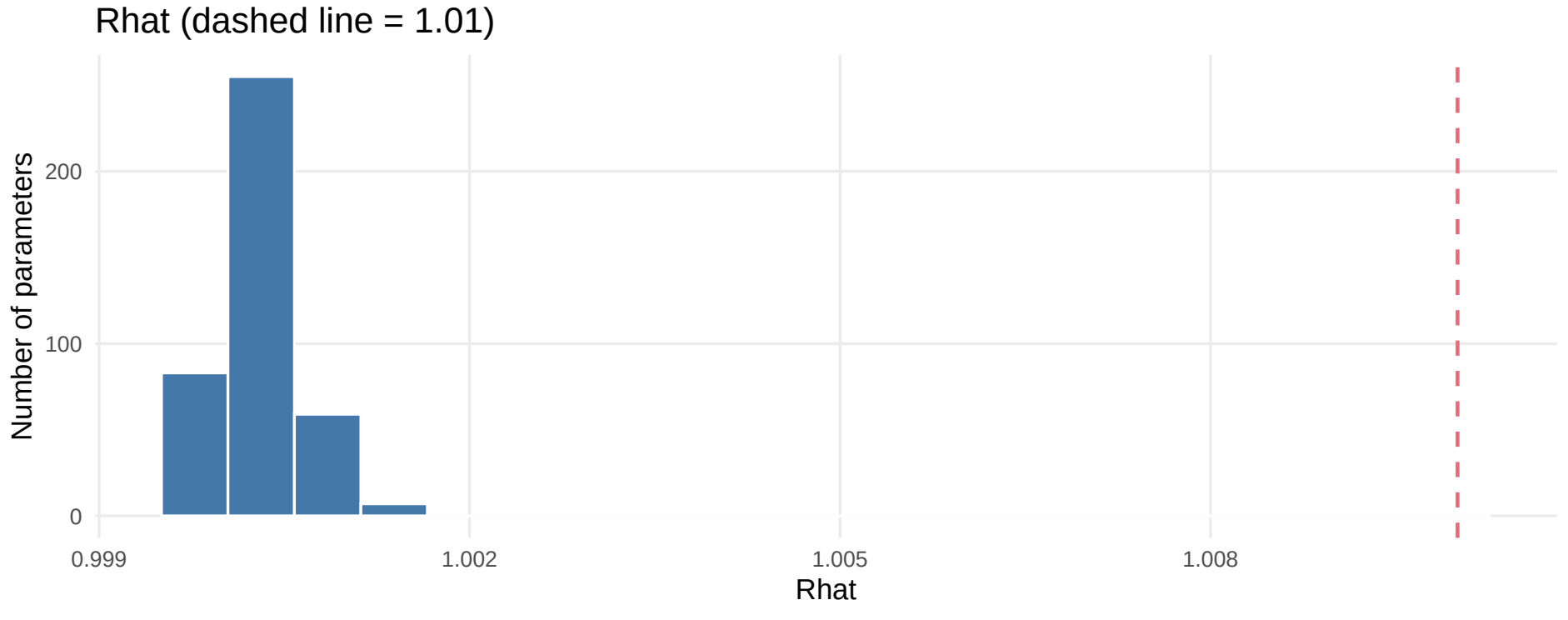

MCMC trace plots (group−level parameters)
mu_alpha
mu_beta
sigma_alpha
sigma_beta
0.25
0.00
−0.25
−0.50
3.25
3.00
2.75
2.1
1.9
1.7
1.5
1.3
1.8
1.6
1.4
1.2
0
500
1000
1500
2000
2500
3000
Chain
1
2
3
4


Trace rank plots (group−level parameters)
mu_alpha
mu_beta
sigma_alpha
sigma_beta
200
150
100
50
0
0
2500
5000
7500
10000
12500
Rank
Chain
1
2
3
4


mu_alpha
mu_beta
sigma_alpha
sigma_beta


ESS (dashed line = 400)
Number of parameters
60
40
20
0
0
10000
20000
Bulk effective sample size


Rhat (dashed line = 1.01)
Number of parameters
200
100
0
0.999
1.002
1.005
1.008
Rhat